PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
LONGIDO DISTRICT EXAMINATION SYNDICATE
FORM TWO MOCK EXAMINATION ,SEPTEMBER -2019
CHEMISTRY

TIME:2.30HRS

INSTRUCTIONS

1.This paper consists sections **A and B**
2.Answer all question in each section
3.Write your examination number on every page of your answer sheet.
4. All writings must be in blue/black ink or ball point pen
5.Phones and calculators are not allowed in the examination room
6.The following constants can be used

Atomic mass; O=16, H=1, S=32, C=12, N=14, Cu=63.5, Cl=35.5, Na=23, K=39, Ca=40, Mg=24.

SECTION A

1. For each of the following items (i-x) choose the correct answer from the given alternatives and write its letter beside the item number.

I. Separation of consistuents of a mixture by fractional distilation is possible when the consitituents in the mixture differ in;

A. Solubility
B. Boiling point
C. Melting point
D. Freezing point

II. The burning material in class A fire is;

A. Metal,papers and plastics
B. Wood, papers and cloth
C. Petrol, kerosene and diesel
D. Sodium potassium and lithium

III. P and Q are atoms of elements in the same period P is in group I and Q is in group II. Which of the following statement is not true of P and Q.

A. P and Q has the same number of energy levels
B. Q has one more proton than P
C. P has one more electron than Q
D. Atomic number of Q is greater than that of P.

IV. Which of the following reaction represents a chemical change.

A. Heating of solid Ammonium chloride in test tube
B. Burning of candle in air
C. Adding solid sodium chloride in water
D. Putting ink on a filter paper

V. Which of the following set of symbols of element stand for a single element

A. ${}^{16}_{7}Z,\ {}^{17}_{8}Z,\ {}^{18}_{9}Z,$
B. ${}^{16}_{7}Z,\ {}^{17}_{8}Z,\ {}^{18}_{9}Z,$
C. ${}^{16}_{8}Z,\ {}^{17}_{8}Z,\ {}^{18}_{8}Z,$

D. ${}^{16}_{7}Z$, ${}^{16}_{9}Z$, ${}^{16}_{8}Z$,

VI. An element A of electronic configuration 2:8:3 combine with an element "B" with electronic configuration of 2:6. The chemical formula of the compound is

A. $B_6\ A_3$
B. $A_3\ B_3$
C. $A_2\ B_3$
D. $A_2\ B$

VII. If a bunsen flame produces much soot which is the correct conclusion?

A. The air holes is closed
B. The burner gas jet is big
C. The air hole is fully opened
D. The gas supply is poor

VIII. When a substance A and B react to produce a new substance C the react A and B are said to be

A. Miscible liquids
B. Undergo chemical change
C. Form mixture
D. Form solution

IX. When a substance is heated and it direct change from solid to gas the process is called

A. Sublimation
B. Freezing
C. Melting
D. Condensation

X. The chemical substance which react easily with oxygen is

A. Flammable
B. Corrosive
C. Oxidant
D. Explosive

2. Match the items in List A with the items in List B by writing its correct letter below the number of corresponding item in the table provided.

LIST A	LIST B
(i)Substance which behave in three states of matter	A.Reducation
(ii)Ability of an atom to gain electrons toward itself	B.J.J.Thomas
(iii)Addition of hydrogen or removal of oxygen from a substance	C.Water
(iv)Plum pudding model	D.Electronegativity
(v) Baking soda	E.A possible explanation of question asked
(vi) Immiscible liquids	F.Oxidation
(vii) Occur when so much water is condensed that the air can not hold	G.$NaHCO_3$

any more	
(viii) Hypothesis formulation	H.Same atomic number different mass number
(ix) Magnetization of Iron	I.Physical change
(x)Isotopes	J.Precipitation
	K.Cooking oil and water
	L.Dalton atomic theory
	M.Condensation
	N.Sodium carbonate(Na_2CO_3)
	O.Electropositivity
	P.Solid, liquid and gas state
	Q.Null hypothesis
	R.Separation of ironss
	S.Existance of an atom with the same atomic number

SECTION B

3. (a) Differentiate between the following give one point in each
   (i)Electrovalency and covalency
   (ii)Electrovalent bond and covalent bond
   (iii)Relative atomic mass and relative molecula mass
   (iv)Isotopes and isotopy
   (v)Solution and suspension

   (b)List down five sources of heat in the laboratory.

4. (a)Give three reasons why air is the mixture
   (b)Write down three differences between physical change and chemical change
   (c) Find the oxidation number of underlined elements in the following compounds.
   (i)$\underline{Cu}SO_4$
   (ii)Na$\underline{Cl}$
   (iii)H_2
   (iv) $H\underline{C}O_3^-$
5. (a)Calculate the molar mass of the following
   (i)$CaCO_3$
   (ii)$NaNO_3$
   (iii)Na_2SO_4
   (iv)$(NH_4)_2SO_4$

   (b) Give the formula for the following compounds
   (i)Carbondioxide
   (ii)Hydrogen sulphide
   (iii)Potassium sulphate
   (iv)Copper II oxide
   (v)Copper I oxide
   (vi)Ammonium hydroxide

(c) Name the following radicals and state their valencies

(i)SO_3^{2-}

(ii)SO_4^{2-}

(iii)NO_3^-

6. Mention two laboratory rules

(b) (i) Why is hydrogen gas is used in filling baloons?

(ii) When an iron bar was left outside for two night its colour changed into red- brown. Give a reason.

(c) State what will be observed when the following simple experiment are performed.

(i) Blue litmus paper is dipped into hydrochloric acid

(ii) A piece of white paper is placed into a non-luminouse flame

(iii) A glowing splint is lowered into a gas jar full of oxygen

(iv) When the mixture of sand and ammonium chloride is heated in a flasks.

7. Magnesium (Mg) has three stable isotopes

$^{24}_{12}Mg$, $^{25}_{12}Mg$ and $^{26}_{12}Mg$

(a) If there is 79% of $^{24}_{12}Mg$ and equal amount of other remaining two isotopes. Calculate the % of $^{25}_{12}Mg$ and $^{26}_{12}Mg$

(b) Calculate the relative atomic mass of Magnesium

8. (a)Show the covalent bond between the water molecules

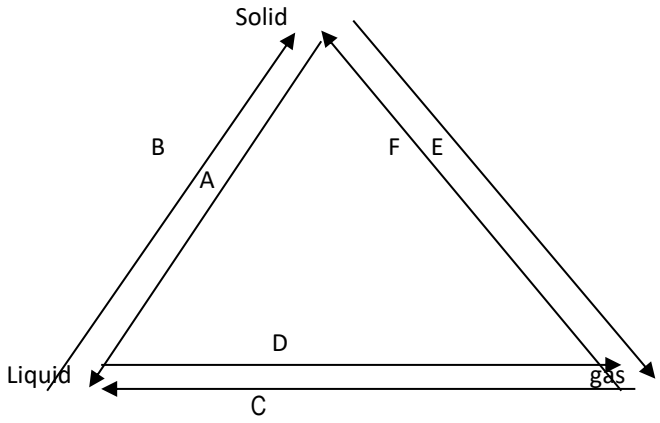
(b)An experiment showed that 13.88g of calcium chloride obtained from the combination of 5g of calcium with 8.8g of Chlorine.

(i) What is empirical formula of calcium chloride

(ii) If the relative atomic mass of sample is 75.5g find its Molecular mass.

9. (a) Define the following term "matter"

(b) Study the diagram below and state the process involved in letter A-F



10. Define the following terms

(i) First aid

(ii) First aid kit

(b)Why we are rendering first aid to sick or injured people? Give three reasons.